Calculus

❶ L'Hôpital's Rule

❷ Extreme Values

❸ Concavity and Inflections, Sketching the Graph of a Function

❹ Extreme-Value Problems

❺ Linear Approximations, Taylor Polynomials

❻ Ideas of Integration

❼ Method of Substitution

❽ Integration by Parts

❾ Inverse Substitutions

❿ Improper Integrals

L'Hôpital's Rule

Intermediate Forms

Table 1: Typical Intermediate Forms of Limit

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We will focus on $\frac{0}{0}, \frac{\infty}{\infty}$; Other forms can sometimes be transformed to these two

L'Hôpital's Rule: Statement

Let f and g be real-valued and differentiable with

$$g' \neq 0 \tag{Denominator' $\neq 0$}$$

on (a, b), $a < b$, $a, b \in \mathbb{R}^\sharp$. Suppose

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The result holds if $x \downarrow a$ is replaced by $x \uparrow b$ and/or ∞ is replaced by $-\infty$.

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L'Hôpital's Rule: Example II

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$$\lim_{x\to\infty} \frac{\ln x}{x^\alpha} = \lim_{x\to\infty} \frac{(\ln x)'}{(x^\alpha)'} = \lim_{x\to\infty} \frac{\frac{1}{x}}{\alpha x^{\alpha-1}} = \lim_{x\to\infty} \frac{1}{\alpha x^\alpha} = 0$$

L'Hôpital's Rule: Example III

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L'Hôpital's Rule: Example III (Cont'd)

Using L'Hôpital's rule again with

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(Denominator' $\neq 0$) $\left(1-\frac{1}{x}+\ln x\right)' = \frac{1+x}{x^2} \neq 0$ on $(1,b)$, $b>1$.

($\frac{0}{0}$) $\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 1+} \ln x = \lim_{x\to 1+} 1-\frac{1}{x}+\ln x = 0$

Hence

$$\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \lim_{x\to 1+} \frac{(\ln x)'}{\left(1-\frac{1}{x}+\ln x\right)'} = \lim_{x\to 1+} \frac{\frac{1}{x}}{\frac{1+x}{x^2}} = \frac{1}{2}$$

L'Hôpital's Rule: Example III (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $\left(1-\frac{1}{x}+\ln x\right)' = \frac{1+x}{x^2} \neq 0$ on $(1,b)$, $b>1$.

($\frac{0}{0}$) $\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 1+} \ln x = \lim_{x\to 1+} 1-\frac{1}{x}+\ln x = 0$

Hence

$$\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \lim_{x\to 1+} \frac{(\ln x)'}{\left(1-\frac{1}{x}+\ln x\right)'} = \lim_{x\to 1+} \frac{\frac{1}{x}}{\frac{1+x}{x^2}} = \frac{1}{2}$$

So

$$\lim_{x\to 1+} \left(\frac{x}{x-1}-\frac{1}{\ln x}\right) = \lim_{x\to 1+} \frac{x\ln x-(x-1)}{(x-1)\ln x} = \lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \frac{1}{2}$$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x \to 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad\left(\text{In the form }\frac{0}{0}\right)$$

and get back to the orginal limit by exponentiation

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad(\text{In the form }\frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad(\text{In the form }\frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)'=2x\neq 0$ on $(0,b)$, $b>0$.

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad(\text{In the form }\frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)'=2x\neq 0$ on $(0,b)$, $b>0$.

($\frac{0}{0}$) $\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}\ln\frac{\sin x}{x}=\lim_{x\to 0}x^2=0$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x} = \lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2} \qquad (\text{In the form } \frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)' = 2x \neq 0$ on $(0, b), b > 0$.

($\frac{0}{0}$) $\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}\ln\frac{\sin x}{x} = \lim_{x\to 0} x^2 = 0$

Hence

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x} = \lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2} \qquad (\text{In the form } \frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)' = 2x \neq 0$ on $(0, b)$, $b > 0$.

($\frac{0}{0}$) $\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}\ln\frac{\sin x}{x} = \lim_{x\to 0}x^2 = 0$

Hence

$$\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad(\text{In the form }\frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)'=2x\neq 0$ on $(0,b)$, $b>0$.

($\frac{0}{0}$) $\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}\ln\frac{\sin x}{x}=\lim_{x\to 0}x^2=0$

Hence

$$\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}=\lim_{x\to 0}\frac{\left(\ln\frac{\sin x}{x}\right)'}{(x^2)'}$$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad(\text{In the form }\frac{0}{0})$$

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Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)'=2x\neq 0$ on $(0,b)$, $b>0$.

($\frac{0}{0}$) $\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}\ln\frac{\sin x}{x}=\lim_{x\to 0}x^2=0$

Hence

$$\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}=\lim_{x\to 0}\frac{\left(\ln\frac{\sin x}{x}\right)'}{(x^2)'}=\lim_{x\to 0}\frac{\frac{x}{\sin x}\frac{(x\cos x-\sin x)}{x^2}}{2x}$$

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

This is of the form 1^∞; Need to be transformed before using L'Hôpital's rule

We consider the logarithm

$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x}=\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}\qquad(\text{In the form }\frac{0}{0})$$

and get back to the orginal limit by exponentiation

Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(x^2)'=2x\neq 0$ on $(0,b)$, $b>0$.

($\frac{0}{0}$) $\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}\ln\frac{\sin x}{x}=\lim_{x\to 0}x^2=0$

Hence

$$\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2}=\lim_{x\to 0}\frac{\left(\ln\frac{\sin x}{x}\right)'}{(x^2)'}=\lim_{x\to 0}\frac{\frac{x}{\sin x}\frac{(x\cos x-\sin x)}{x^2}}{2x}=\lim_{x\to 0}\frac{x\cos x-\sin x}{2x^2\sin x}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{x \cos x - \sin x}{2x^2 \sin x}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{x \cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(x\cos x - \sin x) = \lim_{x\to 0} 2x^2 \sin x = 0$

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(x\cos x - \sin x) = \lim_{x\to 0} 2x^2 \sin x = 0$

Hence

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(x\cos x - \sin x) = \lim_{x\to 0} 2x^2 \sin x = 0$

Hence

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(x\cos x - \sin x) = \lim_{x\to 0} 2x^2 \sin x = 0$

Hence

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x} = \lim_{x\to 0} \frac{(x\cos x - \sin x)'}{(2x^2 \sin x)'}$$

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(x\cos x - \sin x) = \lim_{x\to 0} 2x^2 \sin x = 0$

Hence

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x} = \lim_{x\to 0} \frac{(x\cos x - \sin x)'}{(2x^2 \sin x)'} = \lim_{x\to 0} \frac{-x\sin x}{2x^2 \cos x + 4x \sin x}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(x\cos x - \sin x) = \lim_{x\to 0} 2x^2 \sin x = 0$

Hence

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x} = \lim_{x\to 0} \frac{(x\cos x - \sin x)'}{(2x^2 \sin x)'} = \lim_{x\to 0} \frac{-x\sin x}{2x^2 \cos x + 4x \sin x} = \lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{-\sin x}{2x \cos x + 4 \sin x}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator$' \neq 0$) and ($\frac{0}{0}$):

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

$(\frac{0}{0})$ $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

$(\frac{0}{0})$ $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

So

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

$(\frac{0}{0})$ $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

So

$$\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

$(\frac{0}{0})$ $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

So

$$\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} = \exp\left\{\lim_{x\to 0} \frac{1}{x^2} \ln \frac{\sin x}{x}\right\}$$

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

So

$$\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} = \exp\left\{\lim_{x\to 0} \frac{1}{x^2} \ln \frac{\sin x}{x}\right\} = \exp\left\{\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}\right\}$$

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

($\frac{0}{0}$) $\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 0}(-\sin x) = \lim_{x\to 0}(2x\cos x + 4\sin x) = 0$

Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

So

$$\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} = \exp\left\{\lim_{x\to 0} \frac{1}{x^2} \ln \frac{\sin x}{x}\right\} = \exp\left\{\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}\right\} = \exp\left\{\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}\right\}$$

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Checking the conditions (Denominator$' \neq 0$) and ($\frac{0}{0}$):

(Denominator$' \neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

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(4.3.29)

$$\lim_{x\to 0} (\cos 2x)^{\frac{1}{x^2}}$$

(4.3.30)

$$\lim_{x\to 0+} \frac{\csc x}{\ln x}$$

(4.3.31)

$$\lim_{x\to 0} \frac{\ln \sin \pi x}{\csc \pi x}$$

(4.3.32)

$$\lim_{x\to 0} (1+\tan x)^{\frac{1}{x}}$$

(4.3.33) For twice differentiable f, find

$$\lim_{h\to 0} \frac{f(x+h)-2f(x)+f(x-h)}{h^2}$$

(4.3.34) If f has continuous third derivative, find

$$\lim_{h\to 0} \frac{f(x+3h)-3f(x+h)+3f(x-h)-f(x-3h)}{h^3}$$

Extreme Values

Let $f : I \mapsto \mathbb{R}$.

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The maximum is $f(-1) = 7$, the minimum is $f(2) = -20$.

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Applications

Concavity and Inflections, Sketching the Graph of a Function

Definition of Concavity: Let $f : I \mapsto \mathbb{R}$ be differentiable.

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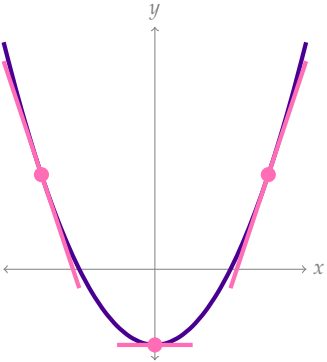
3. If f has an inflection on $x_0 \in I$, then $f''(x_0) = 0$.

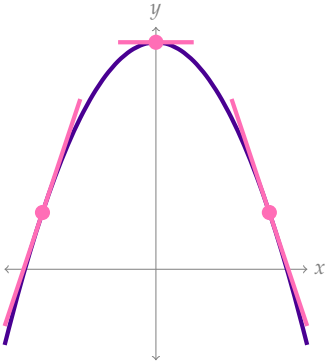


Slopes are increasing

$f''(x) > 0$

"concave up"

tangent line below curve



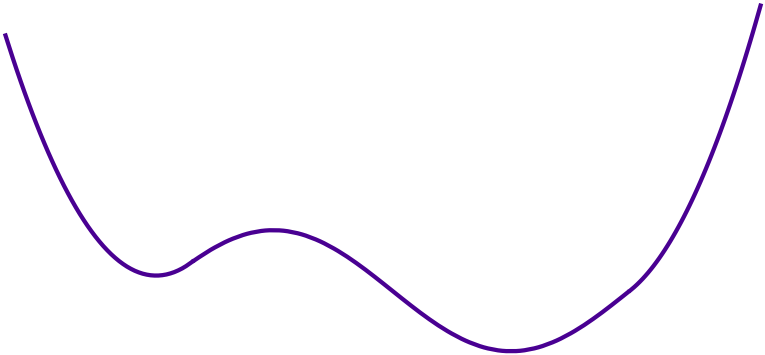
Slopes are decreasing

$f''(x) < 0$

"concave down"

tangent line above curve

Concavity



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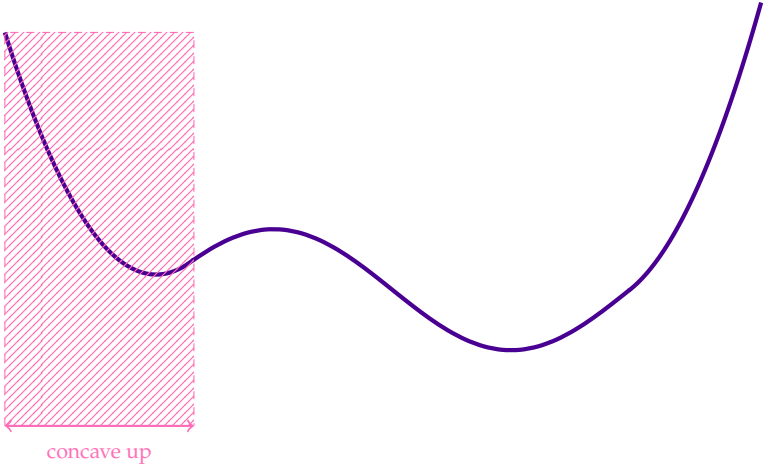

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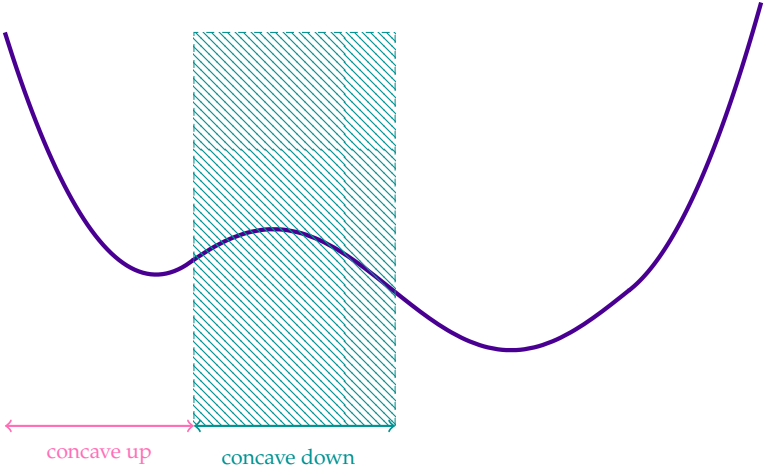

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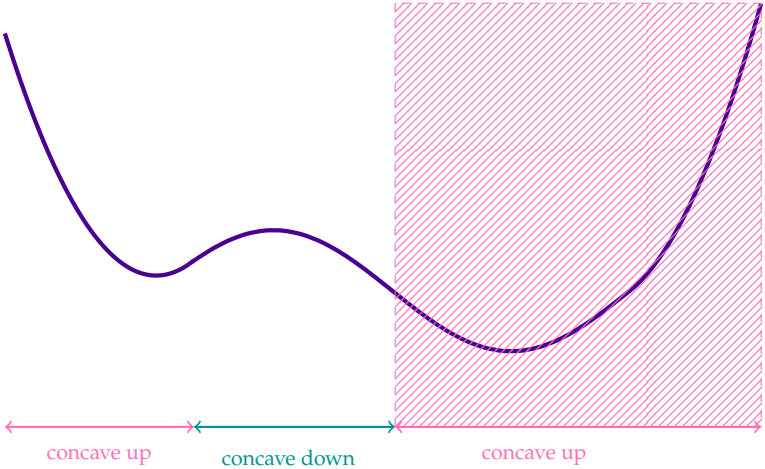
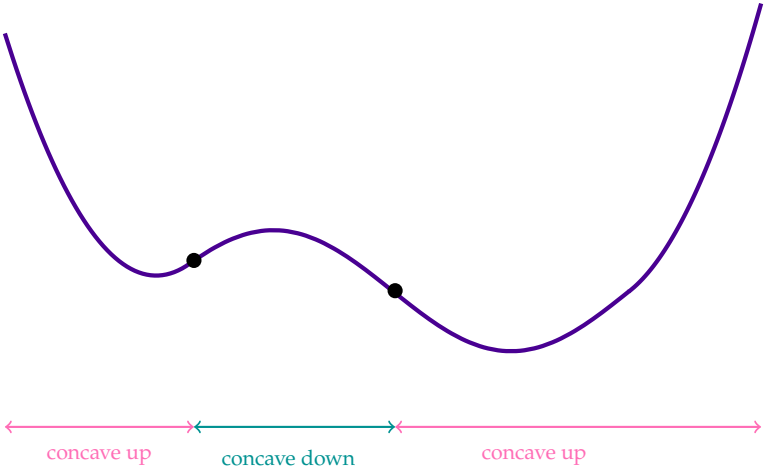
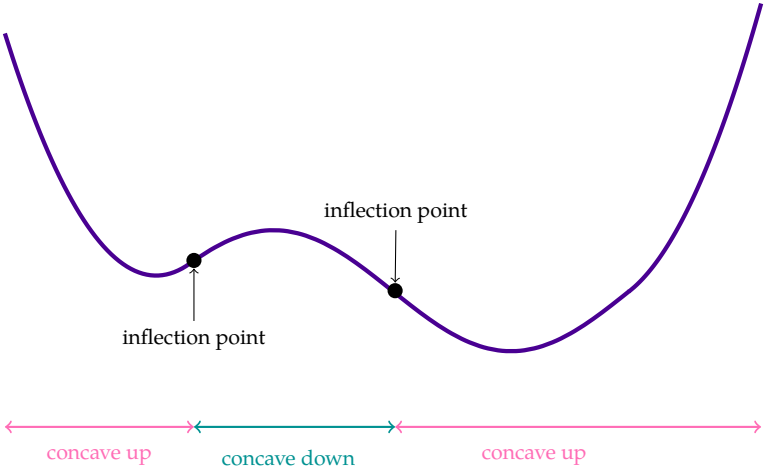
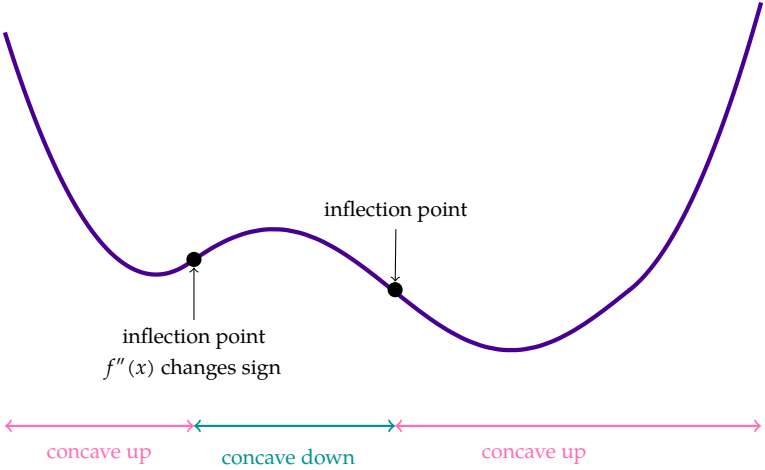


Concavity



Concavity


inflection point
inflection point
concave up
concave down
concave up


inflection point
inflection point
$f''(x)$ changes sign
concave up
concave down
concave up

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2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$,

The First, Second Derivative Tests

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

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The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

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2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$,

The First, Second Derivative Tests

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a,x_0) and $f'(x) < 0$ on (x_0,b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a,x_0) and $f'(x) > 0$ on (x_0,b), then f has a local minimum value at x_0.

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1. If $f'(x) > 0$ on $(b-h,b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h,b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

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2. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

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2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) = 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

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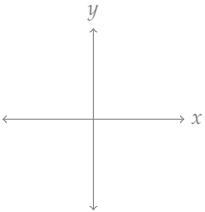
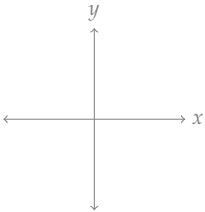
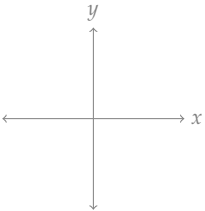
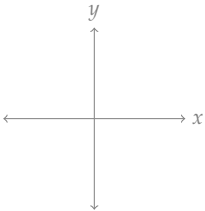
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

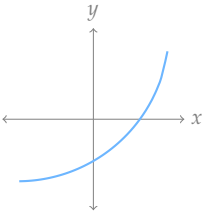
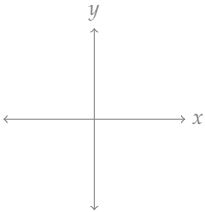
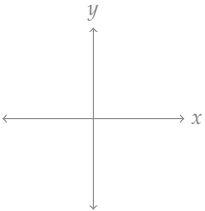
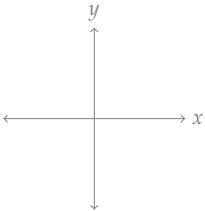
The Second Derivative Test

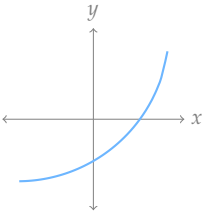
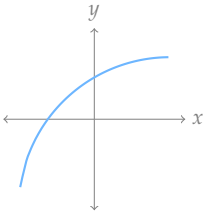
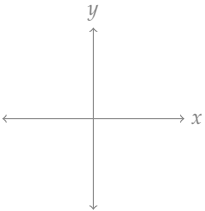
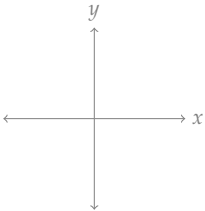
If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

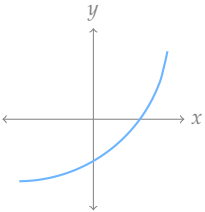
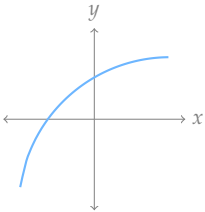
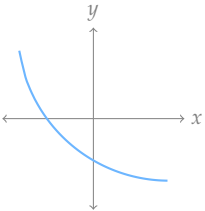
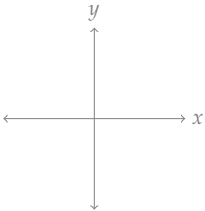
If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

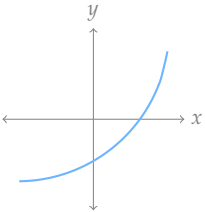
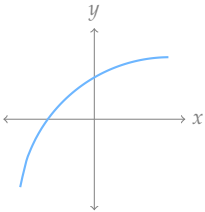
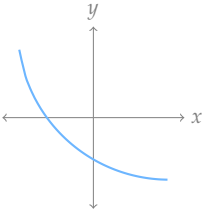
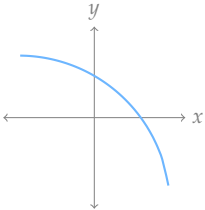
If $f'(x_0) = 0$ and $f''(x_0) = 0$, then no conclusion can be drawn.

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

Example I

Determine the intervals of increase and decrease, the local extreme values, the concavity and the graph of $f(x) = \frac{x}{x^2+3}$.

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$$f'(x) = \frac{3-x^2}{(x^2+3)^2} = \frac{(\sqrt{3}-x)(\sqrt{3}+x)}{(x^2+3)^2},$$

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List the critical and inflection points and decide the sign of f', f'' in each interval:

Example I

Determine the intervals of increase and decrease, the local extreme values, the concavity and the graph of $f(x) = \frac{x}{x^2+3}$.

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List the critical and inflection points and decide the sign of f', f'' in each interval:

	$-$		$-$		$+$		$+$		$-$		$-$	f'
$-$	-3	$+$	$-\sqrt{3}$	$+$	0	$-$	$\sqrt{3}$	$-$	3	$+$		f''

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interval of increase:

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interval of increase: interval that $f' > 0$.

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	$-$		$-$		$+$		$+$		$-$		$-$ f'
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interval of increase: interval that $f' > 0$. $(-\sqrt{3}, \sqrt{3})$.

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	$-$		$-$		$+$		$+$		$-$		$-$	f'
$-$		-3		$+$		$-\sqrt{3}$		$+$		0		$-$	$\sqrt{3}$	$-$	3	$+$	f''

interval of increase: interval that $f' > 0$. $(-\sqrt{3}, \sqrt{3})$.

interval of decrease:

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Determine the intervals of increase and decrease, the local extreme values, the concavity and the graph of $f(x) = \frac{x}{x^2+3}$.

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	$-$		$-$		$+$		$+$		$-$		$-$	f'
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interval of increase: interval that $f' > 0$. $(-\sqrt{3}, \sqrt{3})$.

interval of decrease: interval that $f' < 0$.

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	$-$		$-$		$+$		$+$		$-$		$-$	f'
$-$	-3	$+$	$-\sqrt{3}$	$+$	0	$-$	$\sqrt{3}$	$-$	3	$+$		f''

interval of increase: interval that $f' > 0$. $(-\sqrt{3}, \sqrt{3})$.

interval of decrease: interval that $f' < 0$. $(-\infty, -\sqrt{3})$ or $(\sqrt{3}, \infty)$.

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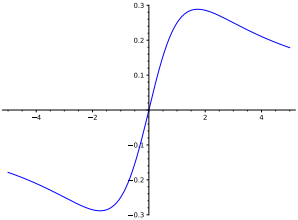

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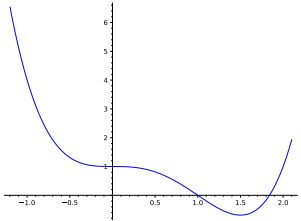

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singular points: $x^2 - 1 = 0$.

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interval of increase: interval that $f' > 0$.

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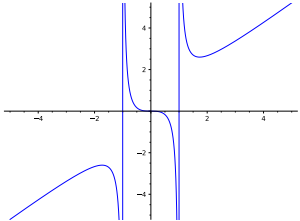
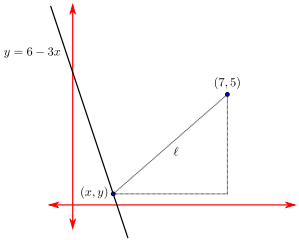


Extreme-Value Problems

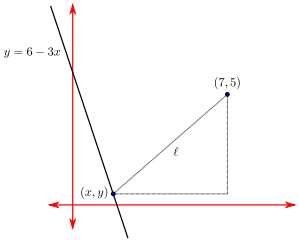
Example I



Find the point on the line $y = 6 - 3x$ that is closest to $(7, 5)$ and the shortest distance.

Review of the Key Theorem

Example I



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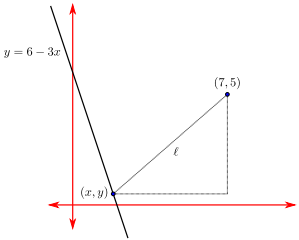

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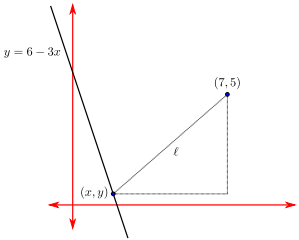

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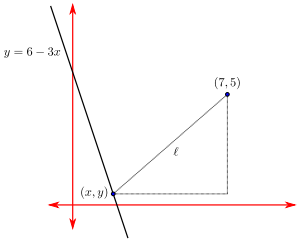

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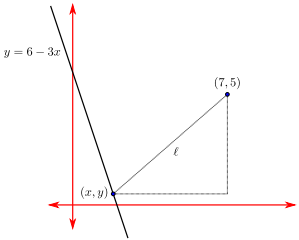

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Compute the derivative of $\ell^2(x)$:

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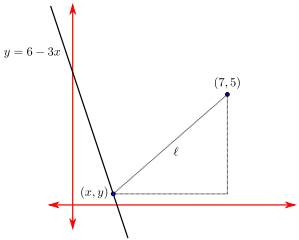

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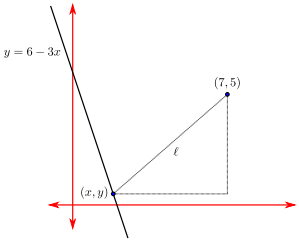

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The minimizer $x^\star$ is the root of $(\ell^2(x))' = 0$, i.e. $x^\star = 1$.

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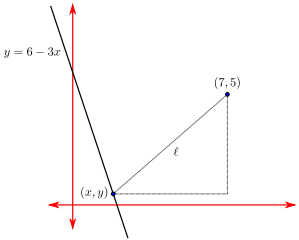

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The minimal distance is $\ell(1) = 2\sqrt{10}$ and occurs at $x = 1$, $y = 6 - 3 = 3$.

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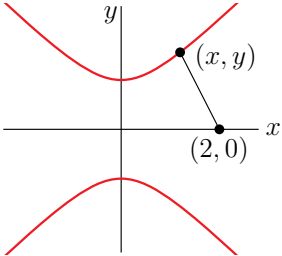

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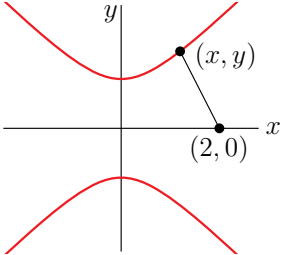

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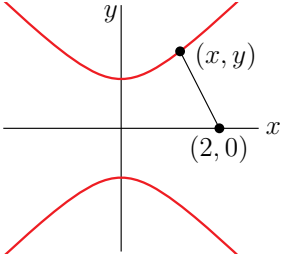

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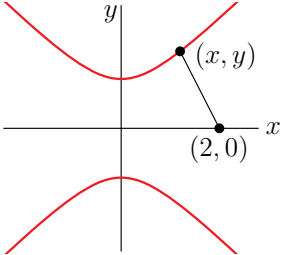

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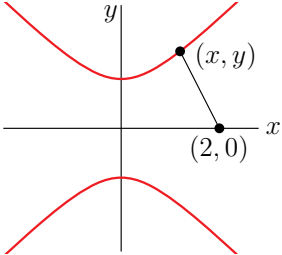

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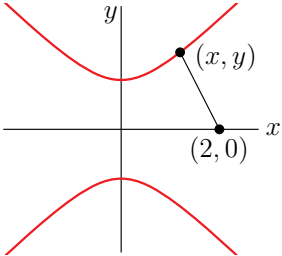

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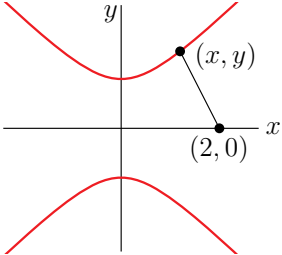

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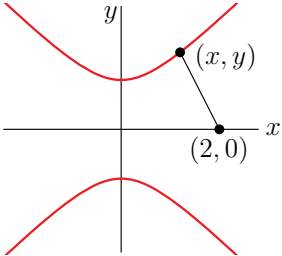

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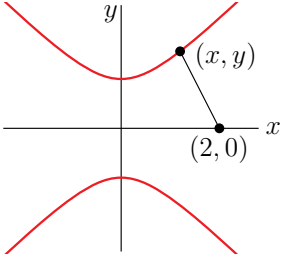

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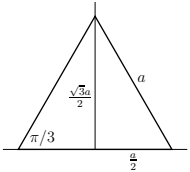

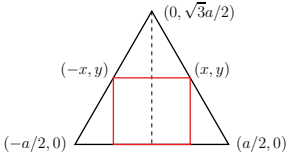

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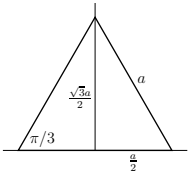

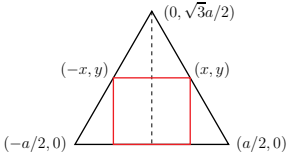

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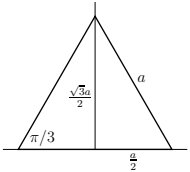

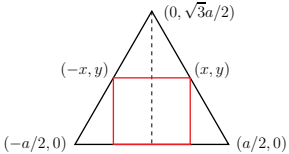

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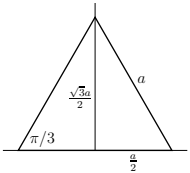

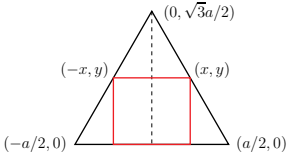

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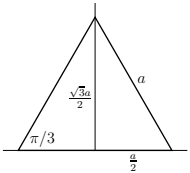

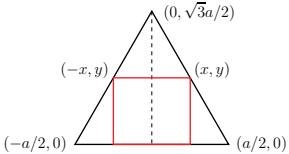

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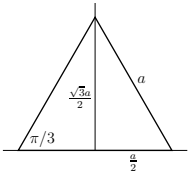

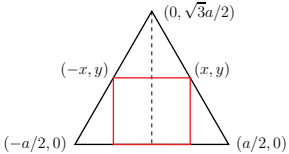

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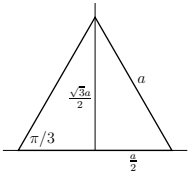

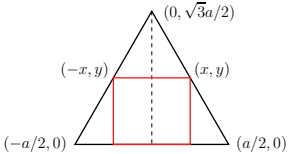

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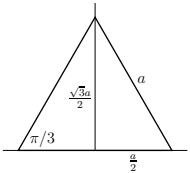

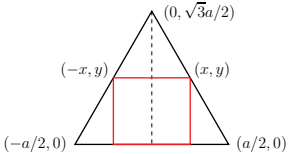

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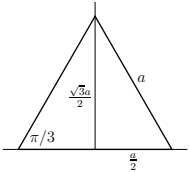

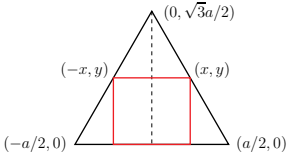

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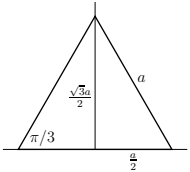

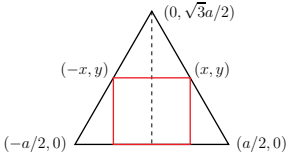

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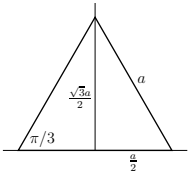

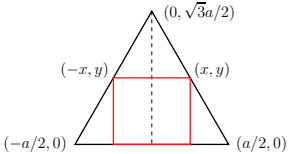

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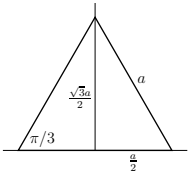

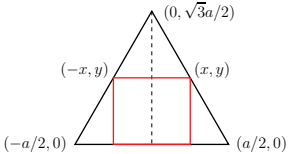

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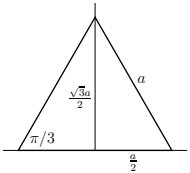
The area of the inscribed rectangle $A = 2xy$; substitute $y = -\sqrt{3}x + \frac{\sqrt{3}}{2}a$, the area of the rectangle $A(x) = 2x\big(-\sqrt{3}x + \frac{\sqrt{3}}{2}a\big) = \sqrt{3}x(a - 2x)$, where $0 \leqslant x \leqslant \frac{a}{2}$.

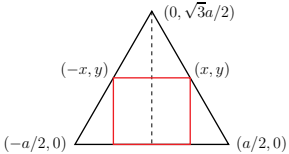
As $x = 0, \frac{a}{2}$, $A(x) = 0$, and $A(\frac{a}{10}) > 0$, so the absolute maximum exists.

Compute the derivative of $A(x)$:

$$A'(x) = \sqrt{3}(x \cdot (-2) + 1 \cdot (a - 2x)) = \sqrt{3}(a - 4x)$$

The minimizer $x^\star$ is the root of $A'(x) = 0$, i.e. $x^\star = \frac{a}{4}$.





Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side a if one side of the rectangle lies on the base of the triangle.

Let (x, y) be a point on the line that connects $(0, \frac{\sqrt{3}}{2}a)$ and $(\frac{a}{2}, 0)$.

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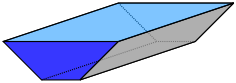
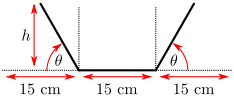
The largest such rectangle has dimensions $2 \times \frac{a}{4} = \frac{a}{2}$ by $-\sqrt{3}\,\frac{a}{4} + \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a$.

Example IV





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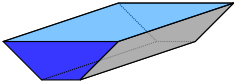

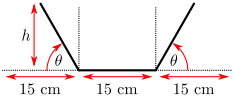

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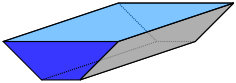

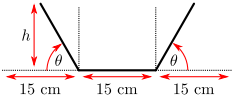

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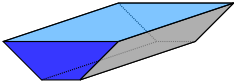

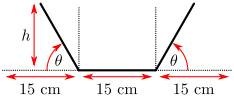

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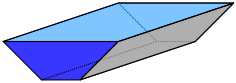

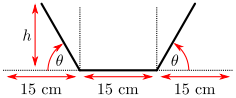

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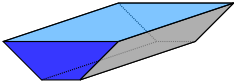

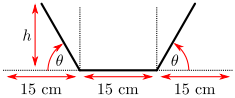

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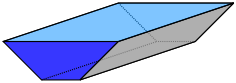

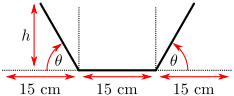

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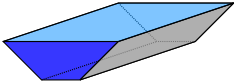

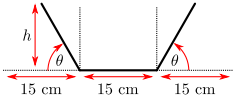

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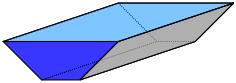

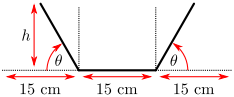

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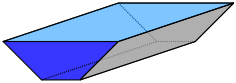

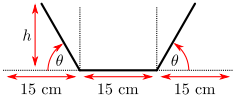

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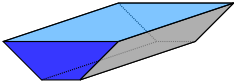

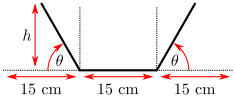

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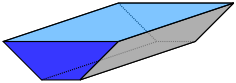

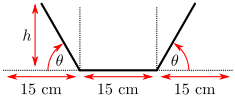

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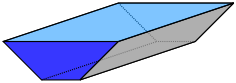

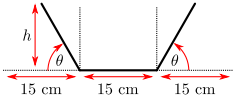

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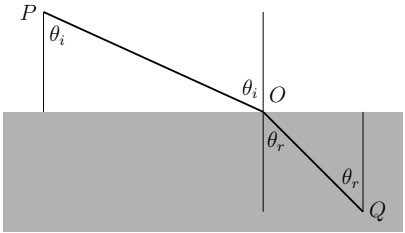
The largest area $A(\frac{\pi}{3}) = 225\sin(\frac{\pi}{3})(\cos(\frac{\pi}{3}) + 1) = 225 \cdot \frac{3\sqrt{3}}{4} \approx 292.28$.

Example V



The Fermat principle states that the path of light travels from P to Q takes minimal time; establish the Snell's law $\frac{\sin\theta_i}{\sin\theta_r} = \frac{c_a}{c_w}$, where c_a, c_w be the speed of light in air and water.

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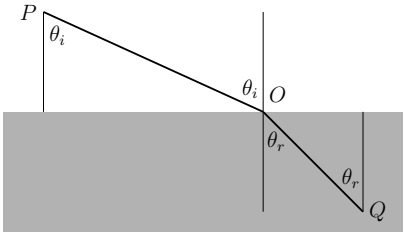

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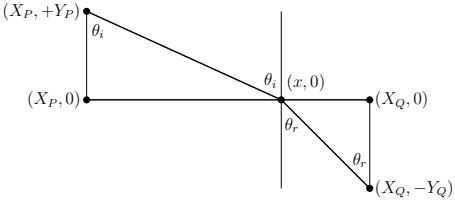

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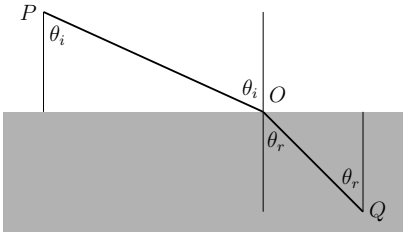

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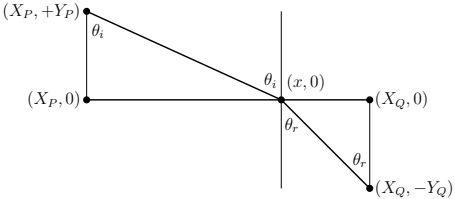

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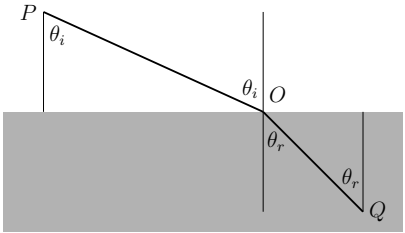

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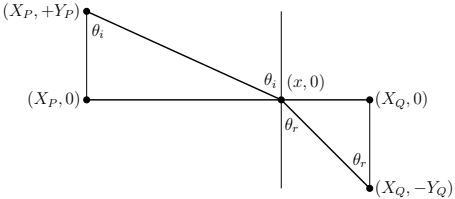

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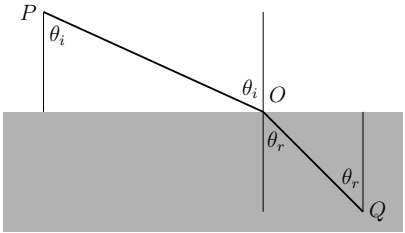

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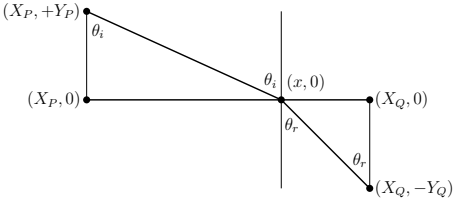

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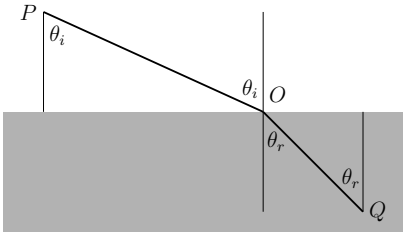

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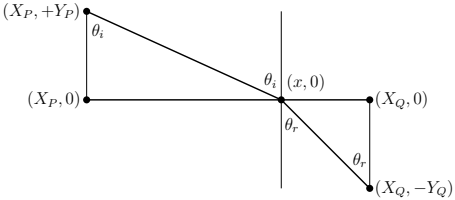

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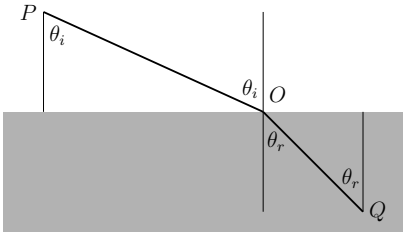

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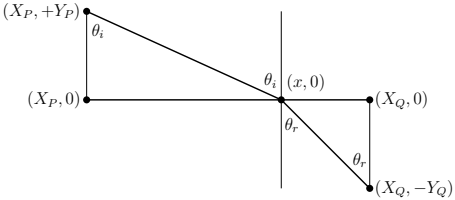

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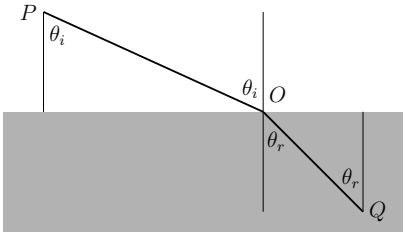

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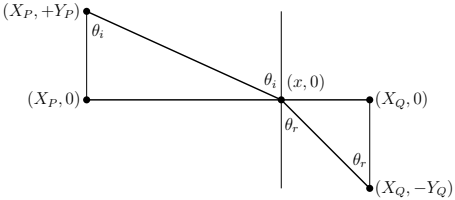

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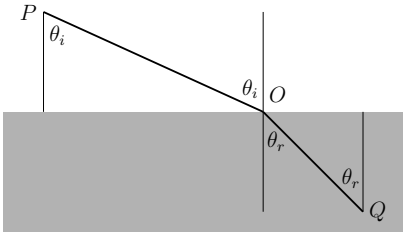

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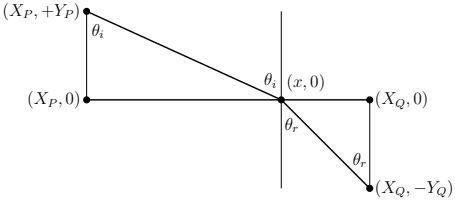

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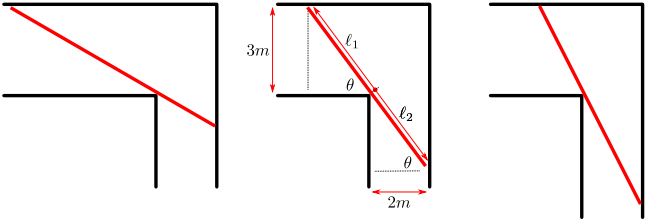

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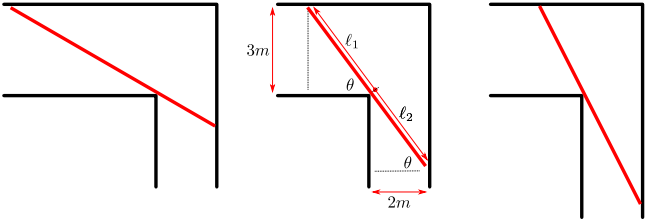

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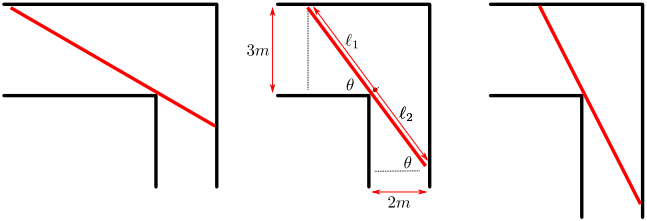

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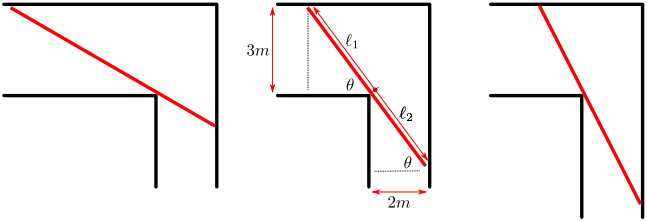

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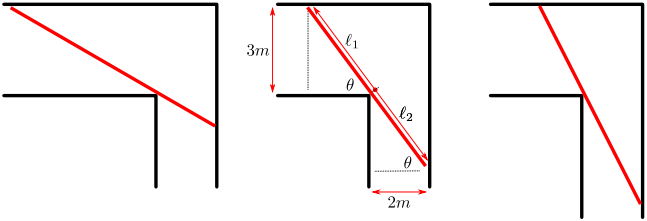

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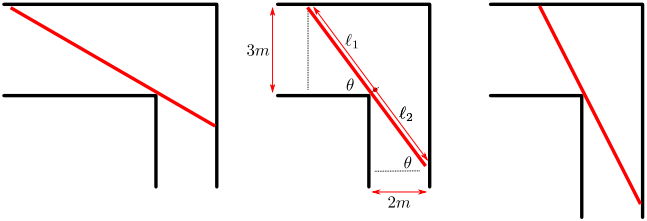

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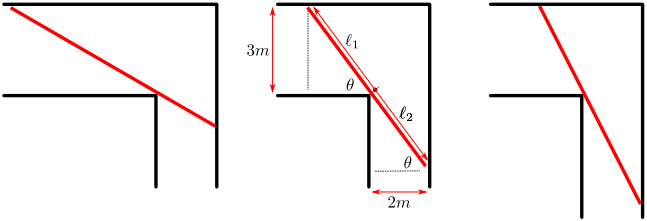

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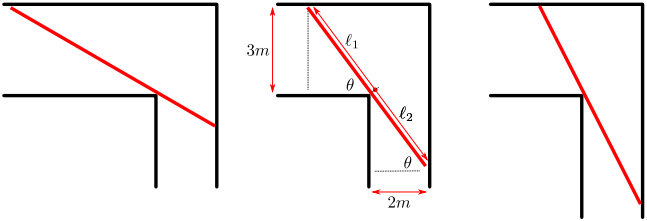

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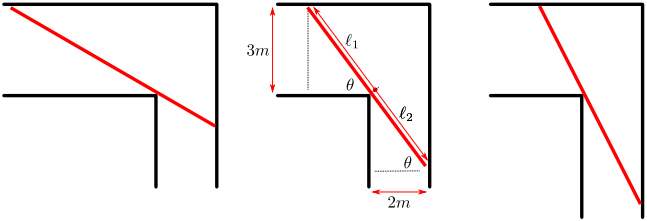

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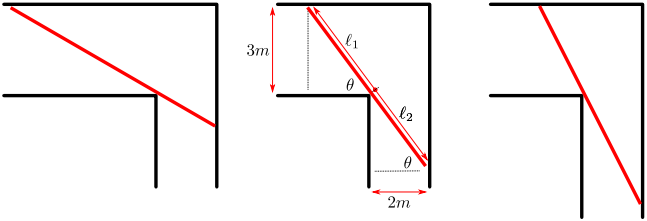

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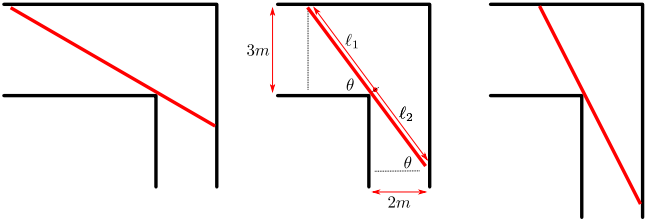

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Linear Approximations, Taylor Polynomials

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$$f(b) = f(a) + f'(a)(b-a) + \frac{f''(a)}{2!}(b-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}$$

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So $g'(c) = 0$ implies $f^{(n+1)}(c) - M = 0 \implies M = f^{(n+1)}(c)$.

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Example: $P_3(x)$ for $f(x) = \ln x$ about $x = e$ is $1 + \frac{1}{e}(x-e) - \frac{1}{2e^2}(x-e)^2 + \frac{1}{3e^3}(x-e)^3$

Example: the Maclaurin polynomial of order 6 for exp is $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$

Example: the Maclaurin polynomial of order 9 for sin is $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!}$

Example: the Maclaurin polynomial of order 8 for cos is $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!}$

Example: the Maclaurin polynomial of order 6 for $\frac{1}{1-x}$ is $1 + x + x^2 + x^3 + x^4 + x^5 + x^6$

Let f be defined on an open interval I, $a, x \in I$, and $f^{(n+1)}$ exists on I, $n \in \mathbb{Z}^+$.

The n-th order Taylor polynomial $P_n(x)$ for f about $x = a$ is defined as

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

The n-th order Taylor polynomial $P_n(x)$ for f about $x = 0$ is also called the Maclaurin polynomial.

$P_1(x)$ for f about $x = a$ is called the linearization of f near $x = a$.

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Example: the Maclaurin polynomial of order 5 for $\ln(1+x)$ is

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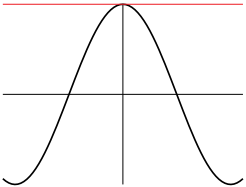
Example: the Maclaurin polynomial of order 9 for sin is $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!}$

Example: the Maclaurin polynomial of order 8 for cos is $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!}$

Example: the Maclaurin polynomial of order 6 for $\frac{1}{1-x}$ is $1 + x + x^2 + x^3 + x^4 + x^5 + x^6$

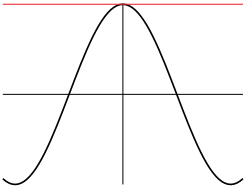
Example: the Maclaurin polynomial of order 5 for $\ln(1+x)$ is $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$

Maclaurin Polynomial Approximation of Cosine



$\cos x \approx 1$

Maclaurin Polynomial Approximation of Cosine



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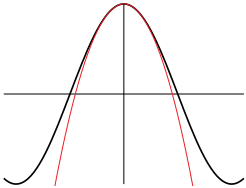
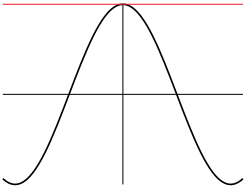


$\cos x \approx 1 - \frac{1}{2!}x^2$

Maclaurin Polynomial Approximation of Cosine



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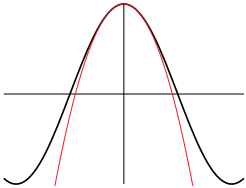
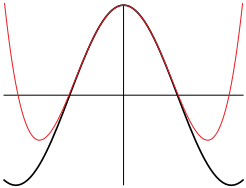
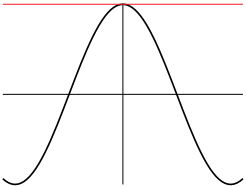


$\cos x \approx 1 - \frac{1}{2!}x^2$

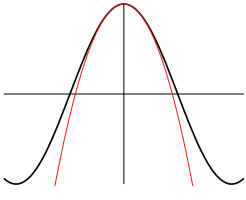


$\cos x \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4$

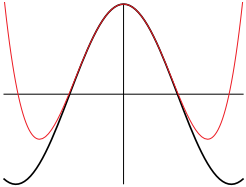
Maclaurin Polynomial Approximation of Cosine



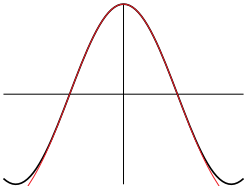
$\cos x \approx 1$



$\cos x \approx 1 - \frac{1}{2!}x^2$

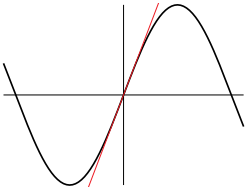


$\cos x \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4$



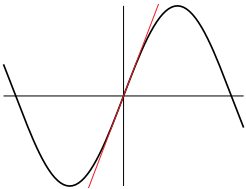
$\cos x \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6$

Maclaurin Polynomial Approximation of Sine

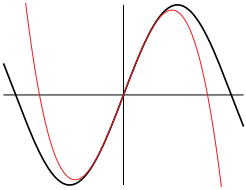


$\sin x \approx x$

Maclaurin Polynomial Approximation of Sine

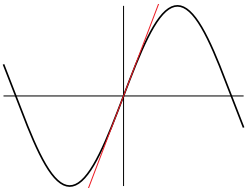


$\sin x \approx x$

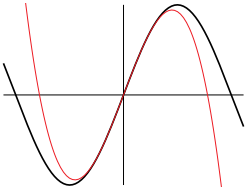


$\sin x \approx x - \frac{1}{3!}x^3$

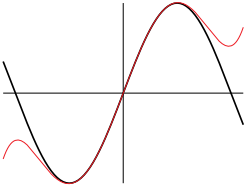
Maclaurin Polynomial Approximation of Sine



$\sin x \approx x$

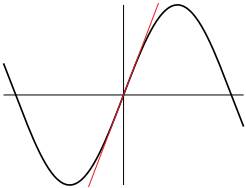


$\sin x \approx x - \frac{1}{3!}x^3$

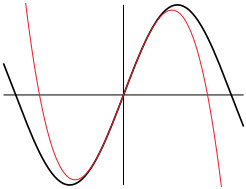


$\sin x \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5$

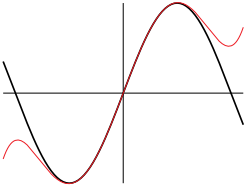
Maclaurin Polynomial Approximation of Sine



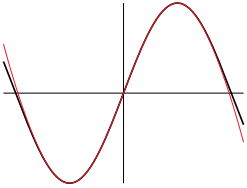
$\sin x \approx x$



$\sin x \approx x - \frac{1}{3!}x^3$

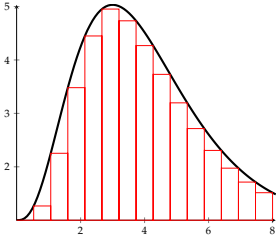


$\sin x \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5$



$\sin x \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7$

Ideas of Integration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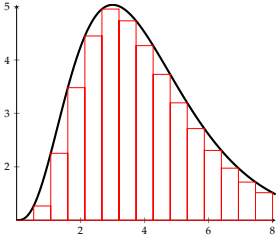
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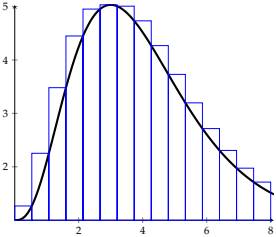
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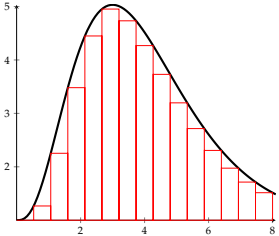
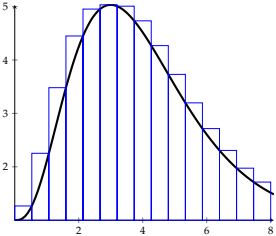
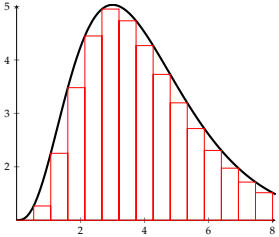
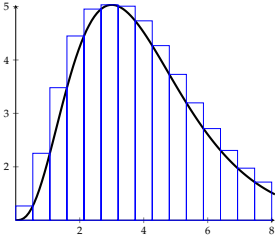




Let $f : [a, b] \mapsto \mathbb{R}$ be continuous, $\mathbb{P} : a = x_0 < x_1 < x_2 < \cdots < x_n = b$ be a partition of $[a, b]$.





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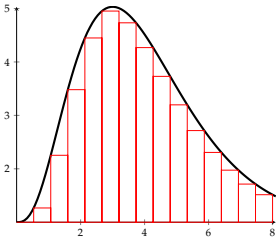
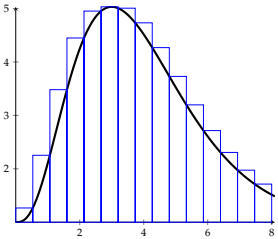
Let $\Delta_k = x_k - x_{k-1}$, $\|\mathbb{P}\| = \max\{|\Delta_k| : 1 \leqslant k \leqslant n\}$, $x_{k-1} \leqslant t_k \leqslant x_k$, $k = 1, 2, \ldots, n$.





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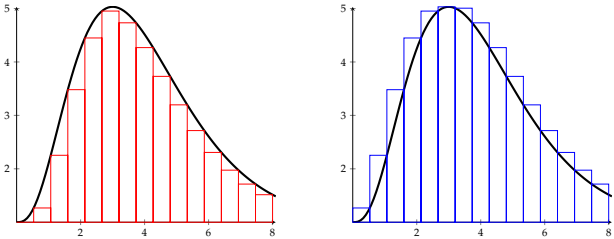
Let $f_k = f(t_k)$, $u_k = \sup\{f(x) : x_{k-1} \leqslant x \leqslant x_k\}$, $l_k = \inf\{f(x) : x_{k-1} \leqslant x \leqslant x_k\}$, $k = 1, 2, \ldots, n$.



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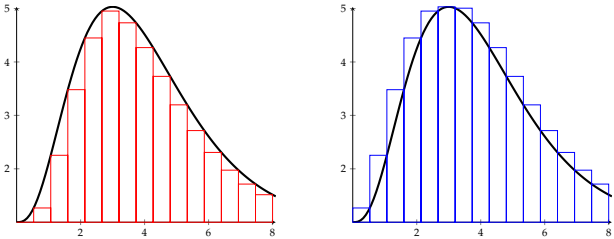
Define the sums



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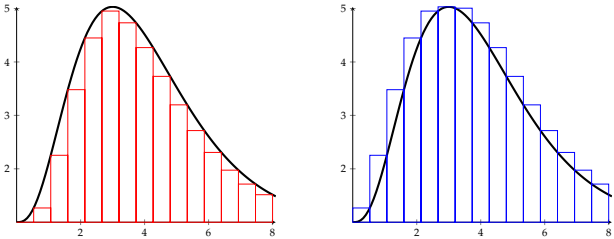
Define the sums $R(f, \mathbb{P}) = \sum_{k=1}^{n} f_k \Delta_k$,



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Let $f_k = f(t_k)$, $u_k = \sup\{f(x) : x_{k-1} \leqslant x \leqslant x_k\}$, $l_k = \inf\{f(x) : x_{k-1} \leqslant x \leqslant x_k\}$, $k = 1, 2, \ldots, n$.

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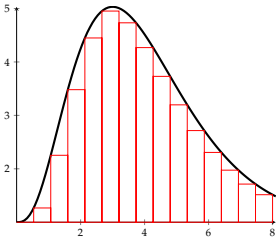

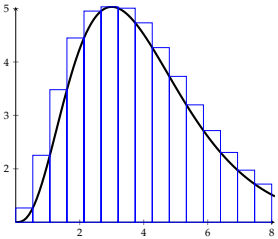

Let $f : [a, b] \mapsto \mathbb{R}$ be continuous, $\mathbb{P} : a = x_0 < x_1 < x_2 < \cdots < x_n = b$ be a partition of $[a, b]$.

Let $\Delta_k = x_k - x_{k-1}$, $\|\mathbb{P}\| = \max\{|\Delta_k| : 1 \leqslant k \leqslant n\}$, $x_{k-1} \leqslant t_k \leqslant x_k$, $k = 1, 2, \ldots, n$.

Let $f_k = f(t_k)$, $u_k = \sup\{f(x) : x_{k-1} \leqslant x \leqslant x_k\}$, $l_k = \inf\{f(x) : x_{k-1} \leqslant x \leqslant x_k\}$, $k = 1, 2, \ldots, n$.

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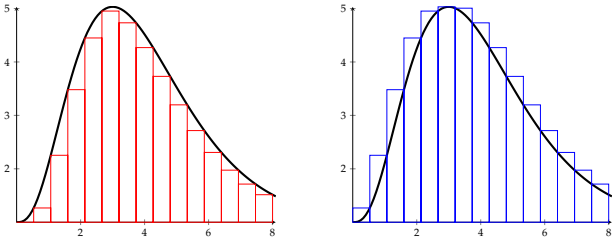

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A priori $L(f, \mathbb{P}) \leqslant R(f, \mathbb{P}) \leqslant U(f, \mathbb{P})$; Define

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\substack{n\to\infty \\ \|\mathbb{P}\|\to 0}} R(f, \mathbb{P})$$

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From FTC, $\int_0^{\frac{\pi}{2}} \cos x \, dx = \sin\frac{\pi}{2} - \sin 0 = 1$!

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Method of Substitution

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$\int \frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x$. Let $u = x-1$, then $\mathrm{d}u = \mathrm{d}x$, the integral becomes $\int \frac{1}{\sqrt{5-u^2}}\,\mathrm{d}u = \arcsin\frac{u}{\sqrt{5}} + C$

Example: evaluate $\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x$.

Let $u = e^x$, then $\mathrm{d}u = e^x\,\mathrm{d}x$. So $\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x = \int \frac{1}{\sqrt{2-u^2}}\,\mathrm{d}u = \arcsin\frac{u}{\sqrt{2}} + C = \arcsin\frac{e^x}{\sqrt{2}} + C$

Example: evaluate $\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x$.

$\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x = \int \frac{1}{e^x\sqrt{1-e^{-2x}}}\,\mathrm{d}x = \int \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x$. Let $u = e^{-x}$, then $\mathrm{d}u = -e^{-x}\mathrm{d}x$, the integral becomes $-\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u$ $= -\arcsin u + C = -\arcsin e^{-x} + C$.

Example: evaluate $\int \frac{1}{x^2+4x+5}\,\mathrm{d}x$.

$\int \frac{1}{x^2+4x+5}\,\mathrm{d}x = \int \frac{1}{(x+2)^2+1}\,\mathrm{d}x$. Let $u = x+2$, then $\mathrm{d}u = \mathrm{d}x$, the integral becomes $\int \frac{1}{u^2+1}\,\mathrm{d}u = \arctan u + C$ $= \arctan(x+2) + C$.

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$\int \frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x$. Let $u = x-1$, then $\mathrm{d}u = \mathrm{d}x$, the integral becomes $\int \frac{1}{\sqrt{5-u^2}}\,\mathrm{d}u = \arcsin\frac{u}{\sqrt{5}} + C$ $= \arcsin\frac{x-1}{\sqrt{5}} + C$.

Example: evaluate $\int \tan x \, \mathrm{d}x$.

Example: evaluate $\int \tan x \, \mathrm{d}x$.

$\int \tan x \, \mathrm{d}x = \int \frac{\sin x}{\cos x} \, \mathrm{d}x.$

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$\int \tan x \, \mathrm{d}x = \int \frac{\sin x}{\cos x} \, \mathrm{d}x$. Let $u = \cos x$, then $\mathrm{d}u = -\sin x \, \mathrm{d}x$.

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$\int \tan x \, \mathrm{d}x = \int \frac{\sin x}{\cos x} \, \mathrm{d}x$. Let $u = \cos x$, then $\mathrm{d}u = -\sin x \, \mathrm{d}x$. So $\int \frac{\sin x}{\cos x} \, \mathrm{d}x = \int \frac{-1}{u} \, \mathrm{d}u$

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Example: evaluate $\int \cos^5 ax \, dx$.

Example: evaluate $\int \tan x \, \mathrm{d}x$.

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Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$.

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$\int \cos^5 ax \, \mathrm{d}x = \int (1 - \sin^2 ax)^2 \cos ax \, \mathrm{d}x$. Let $u = \sin ax$, then $\mathrm{d}u = a \cos ax \, \mathrm{d}x$, so $\int (1 - \sin^2 ax)^2 \cos ax \, \mathrm{d}x$

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Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du$

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$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du$

Example: evaluate $\int \tan x \, dx$.

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Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du = \frac{1}{a}(u - \frac{2}{3}u^3 + \frac{1}{5}u^5) + C$

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Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du = \frac{1}{a}(u - \frac{2}{3}u^3 + \frac{1}{5}u^5) + C = \frac{1}{a}(\sin ax - \frac{2}{3}\sin^3 ax + \frac{1}{5}\sin^5 ax) + C$

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$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$. Let $u = \cos x$, then $du = -\sin x \, dx$. So $\int \frac{\sin x}{\cos x} \, dx = \int \frac{-1}{u} \, du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

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Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du = \frac{1}{a}(u - \frac{2}{3}u^3 + \frac{1}{5}u^5) + C = \frac{1}{a}(\sin ax - \frac{2}{3}\sin^3 ax + \frac{1}{5}\sin^5 ax) + C$

Example: evaluate $\int \cos^4 ax \, dx$.

Example: evaluate $\int \tan x \, dx$.

$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$. Let $u = \cos x$, then $du = -\sin x \, dx$. So $\int \frac{\sin x}{\cos x} \, dx = \int \frac{-1}{u} \, du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

Example: evaluate $\int \sec x \, dx$.

$\int \sec x \, dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} \, dx$. Let $u = \sec x + \tan x$, then $du = (\sec^2 x + \sec x \tan x) \, dx$, so $\int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} \, dx = \int \frac{1}{u} \, du = \ln|u| + C = \ln|\sec x + \tan x| + C$.

Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du = \frac{1}{a}(u - \frac{2}{3}u^3 + \frac{1}{5}u^5) + C = \frac{1}{a}(\sin ax - \frac{2}{3}\sin^3 ax + \frac{1}{5}\sin^5 ax) + C$

Example: evaluate $\int \cos^4 ax \, dx$.

$\int \cos^4 ax \, dx$

Example: evaluate $\int \tan x \, dx$.

$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$. Let $u = \cos x$, then $du = -\sin x \, dx$. So $\int \frac{\sin x}{\cos x} \, dx = \int \frac{-1}{u} \, du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

Example: evaluate $\int \sec x \, dx$.

$\int \sec x \, dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} \, dx$. Let $u = \sec x + \tan x$, then $du = (\sec^2 x + \sec x \tan x) \, dx$, so $\int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} \, dx = \int \frac{1}{u} \, du = \ln|u| + C = \ln|\sec x + \tan x| + C$.

Example: evaluate $\int \cos^5 ax \, dx$.

$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du = \frac{1}{a}(u - \frac{2}{3}u^3 + \frac{1}{5}u^5) + C = \frac{1}{a}(\sin ax - \frac{2}{3}\sin^3 ax + \frac{1}{5}\sin^5 ax) + C$

Example: evaluate $\int \cos^4 ax \, dx$.

$\int \cos^4 ax \, dx = \int \left(\frac{1 + \cos 2ax}{2}\right)^2 dx$

Example: evaluate $\int \tan x \, dx$.

$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$. Let $u = \cos x$, then $du = -\sin x \, dx$. So $\int \frac{\sin x}{\cos x} \, dx = \int \frac{-1}{u} \, du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

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Example: evaluate $\int \cos^4 ax \, dx$.

$\int \cos^4 ax \, dx = \int (\frac{1 + \cos 2ax}{2})^2 \, dx = \frac{1}{4} \int (1 + 2\cos 2ax + \cos^2 2ax) \, dx$

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$\int \cos^4 ax \, \mathrm{d}x = \int (\frac{1+\cos 2ax}{2})^2 \, \mathrm{d}x = \frac{1}{4} \int (1 + 2\cos 2ax + \cos^2 2ax) \, \mathrm{d}x = \frac{1}{4} \int (1 + 2\cos 2ax + \frac{1+\cos 4ax}{2}) \, \mathrm{d}x = \int (\frac{3}{8} + \frac{1}{2}\cos 2ax + \frac{1}{8}\cos 4ax) \, \mathrm{d}x$

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Note: your answers may appear different from the ones here but may still be correct, because of the constant C.

Exercises

Note: your answers may appear different from the ones here but may still be correct, because of the constant C.

$$\int e^{2x} \sin e^{2x} \, \mathrm{d}x$$

Note: your answers may appear different from the ones here but may still be correct, because of the constant C.

$$\int e^{2x} \sin e^{2x} \, \mathrm{d}x = -\frac{1}{2} \cos e^{2x} + C$$

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$$\int e^{2x} \sin e^{2x} \, \mathrm{d}x = -\frac{1}{2} \cos e^{2x} + C$$

$$\int x e^{-\frac{x^2}{2}} \, \mathrm{d}x$$

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$$\int x^2 2^{x^3+1} \, \mathrm{d}x$$

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$$\int \sin^4 x\cos^5 x\,\mathrm{d}x = \frac{\sin^5 x}{5} - \frac{2\sin^7 x}{7} + \frac{\sin^9 x}{9} + C$$

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$$\int \sin ax\sin bx\,dx = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)} + C$$

$$\int \sin ax\cos bx\,dx$$

Note: your answers may appear different from the ones here but may still be correct, because of the constant C.

$$\int e^{2x}\sin e^{2x}\,\mathrm{d}x = -\frac{1}{2}\cos e^{2x} + C$$

$$\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = -e^{-\frac{x^2}{2}} + C$$

$$\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = -2\cos\sqrt{x} + C$$

$$\int x^2 2^{x^3+1}\,\mathrm{d}x = \frac{2^{x^3+1}}{3\ln 2} + C$$

$$\int \frac{\ln x}{x}\,\mathrm{d}x = \frac{1}{2}(\ln x)^2 + C$$

$$\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x = \sqrt{x^2+2x+3} + C$$

$$\int \frac{x^2}{\sqrt{2+x^6}}\,\mathrm{d}x = \frac{1}{3\sqrt{2}}\arctan\frac{x^3}{\sqrt{2}} + C$$

$$\int \frac{1}{e^x+e^{-x}}\,\mathrm{d}x = \arctan e^x + C$$

$$\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x = -\sqrt{1-x^2} + \arcsin x + C$$

$$\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x = \arcsin\tan x + C$$

$$\int \sin^4 x\cos^5 x\,\mathrm{d}x = \frac{\sin^5 x}{5} - \frac{2\sin^7 x}{7} + \frac{\sin^9 x}{9} + C$$

$$\int \sin^2 x\cos^2 x\,\mathrm{d}x = \frac{x}{8} - \frac{\sin 4x}{32} + C$$

$$\int \sin^{-\frac{2}{3}} x\cos^3 x\,\mathrm{d}x = 3\sin^{\frac{1}{3}} x - \frac{3}{7}\sin^{\frac{7}{3}} x + C$$

$$\int \frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x = \frac{\sin^4(\ln x)}{4} - \frac{\sin^6(\ln x)}{6} + C$$

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Integration by Parts

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Example: evaluate $\int x^2 \sin x \, dx$.

Example: evaluate $\int x^2 \sin x \, \mathrm{d}x$.

Let $u = x^2$, then $\mathrm{d}u = 2x \, \mathrm{d}x$.

Example: evaluate $\int x^2 \sin x \, \mathrm{d}x$.

Let $u = x^2$, then $\mathrm{d}u = 2x \, \mathrm{d}x$. Let $\mathrm{d}v = \sin x \, \mathrm{d}x$, then $v = -\cos x$.

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Exercises

$$\int (x+3)e^{2x}\,\mathrm{d}x$$

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Inverse Substitutions

Improper Integrals

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From $\Gamma(x+1) = x\,\Gamma(x)$, $\Gamma(\frac{3}{2}) = \Gamma(\frac{1}{2}+1) = \frac{1}{2}\,\Gamma(\frac{1}{2}) = \frac{1}{2}\sqrt{\pi}$

Example: given that $\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$ and $a > 0$, evaluate $\int_0^\infty t\,e^{-a\,t^2}\,\mathrm{d}t$ and $\int_0^\infty t^2\,e^{-a\,t^2}\,\mathrm{d}t$.

Let $u = \sqrt{a}\,t$, then $t = \frac{u}{\sqrt{a}}$, $\mathrm{d}u = \sqrt{a}\,\mathrm{d}t$. So $\int_0^\infty t\,e^{-a\,t^2}\,\mathrm{d}t = \int_0^\infty \frac{u}{\sqrt{a}}\,e^{-u^2}\,\frac{1}{\sqrt{a}}\,\mathrm{d}u = \frac{1}{a}\int_0^\infty u\,e^{-u^2}\,\mathrm{d}u$. Let $x = u^2$, then $\mathrm{d}x = 2u\,\mathrm{d}u$. So $\int_0^\infty t\,e^{-a\,t^2}\,\mathrm{d}t = \frac{1}{a}\int_0^\infty u\,e^{-u^2}\,\mathrm{d}u = \frac{1}{2a}\int_0^\infty e^{-x}\,\mathrm{d}x = \frac{1}{2a}$.

Let $w = \sqrt{a}\,t$, then $t = \frac{w}{\sqrt{a}}$, $\mathrm{d}w = \sqrt{a}\,\mathrm{d}t$. So $\int_0^\infty t^2\,e^{-a\,t^2}\,\mathrm{d}t = \int_0^\infty \frac{w^2}{a}\,e^{-w^2}\,\frac{1}{\sqrt{a}}\,\mathrm{d}w = \frac{1}{a^{\frac{3}{2}}}\int_0^\infty w^2\,e^{-w^2}\,\mathrm{d}w$. Let $u = -\frac{1}{2}w$, then $\mathrm{d}u = -\frac{1}{2}\,\mathrm{d}w$. Let $\mathrm{d}v = -2\,w\,e^{-w^2}\,\mathrm{d}w$, then $v = e^{-w^2}$. So $\int_0^\infty t^2\,e^{-a\,t^2}\,\mathrm{d}t = \frac{1}{a^{\frac{3}{2}}}\int_0^\infty w^2\,e^{-w^2}\,\mathrm{d}w$
$= \frac{1}{a^{\frac{3}{2}}}\left(-\frac{1}{2}w\cdot e^{-w^2}\Big|_0^\infty + \frac{1}{2}\int_0^\infty e^{-w^2}\,\mathrm{d}w \right) = \frac{1}{a^{\frac{3}{2}}}\cdot\frac{1}{2}\cdot\frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{4a^{\frac{3}{2}}}$.